NETWORK TRAFFIC FORENSICS

Malware Delivery, Host Identification & Incident Analysis with Wireshark

Technical investigation | Pedro Gonçalves

UFCD 9192 · DFIR · Network Forensics · Incident Response



Packet capture > Evidence extraction > Indicator correlation

1. Investigation overview

This case study documents the analysis of a Windows malware incident using a supplied PCAP. The workflow covers identification of the affected endpoint, examination of HTTP delivery traffic, executable extraction, hash analysis, and investigation of suspected post-infection communications. Findings are based on the supplied laboratory report; the original PCAP was not provided for independent revalidation.

2. Investigation objectives

Identify the affected endpoint IP, MAC address and hostname; determine the active Windows account; locate the delivery server; inspect the downloaded executable and its SHA-256; examine malware classification and post-infection indicators; and preserve supporting packet-analysis screenshots.

3. Analysis environment and methods

Tools documented: Wireshark, Kali Linux, HTTP Follow TCP Stream, Export Objects (HTTP), packet filtering, and SHA-256 analysis. Relevant protocol families include DHCP/BOOTP, NBNS, LLMNR, SMB, Kerberos and HTTP.

4. Host identification

The DHCP hostname field (bootp.option.hostname) and NBNS/LLMNR/SMB traffic were used to correlate the affected Windows machine. Reported hostname: BANGKOK-8AC2-PC; IPv4: 10.0.76.109; MAC: 78:2b:cb:d4:a5:fe.

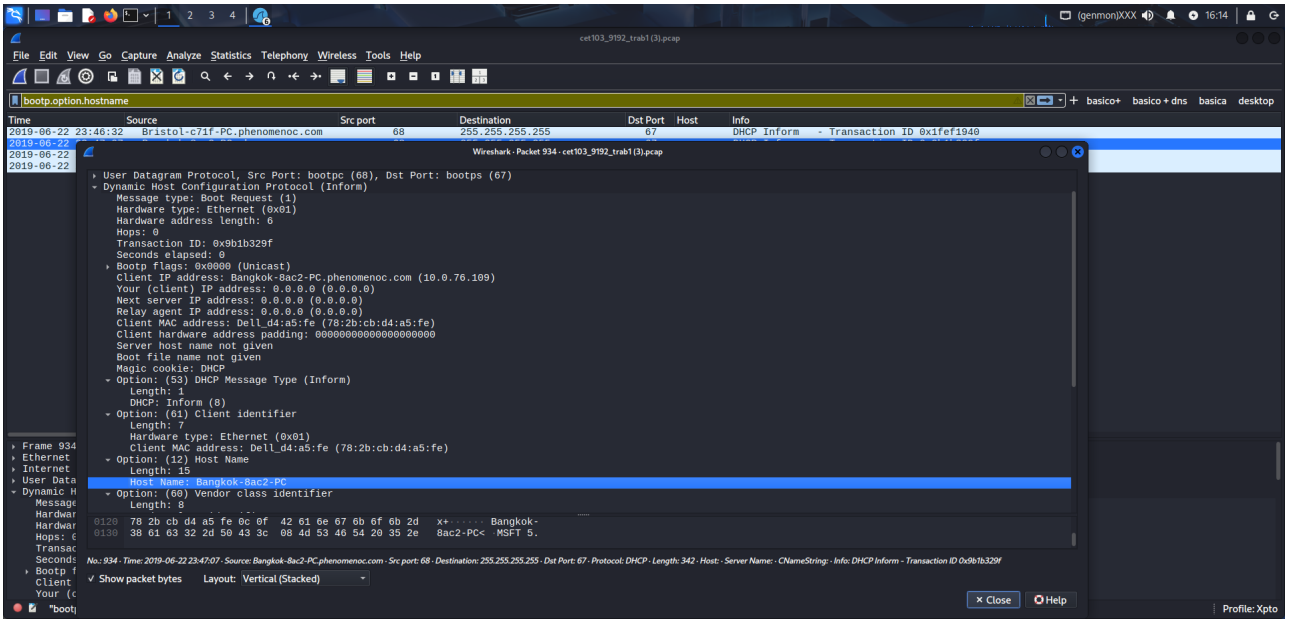


Figure 1 — Wireshark DHCP hostname evidence

5. Windows account attribution

Kerberos client-name fields (kerberos.CNameString) were examined to identify the logged-on account. The source report records edris.haight.

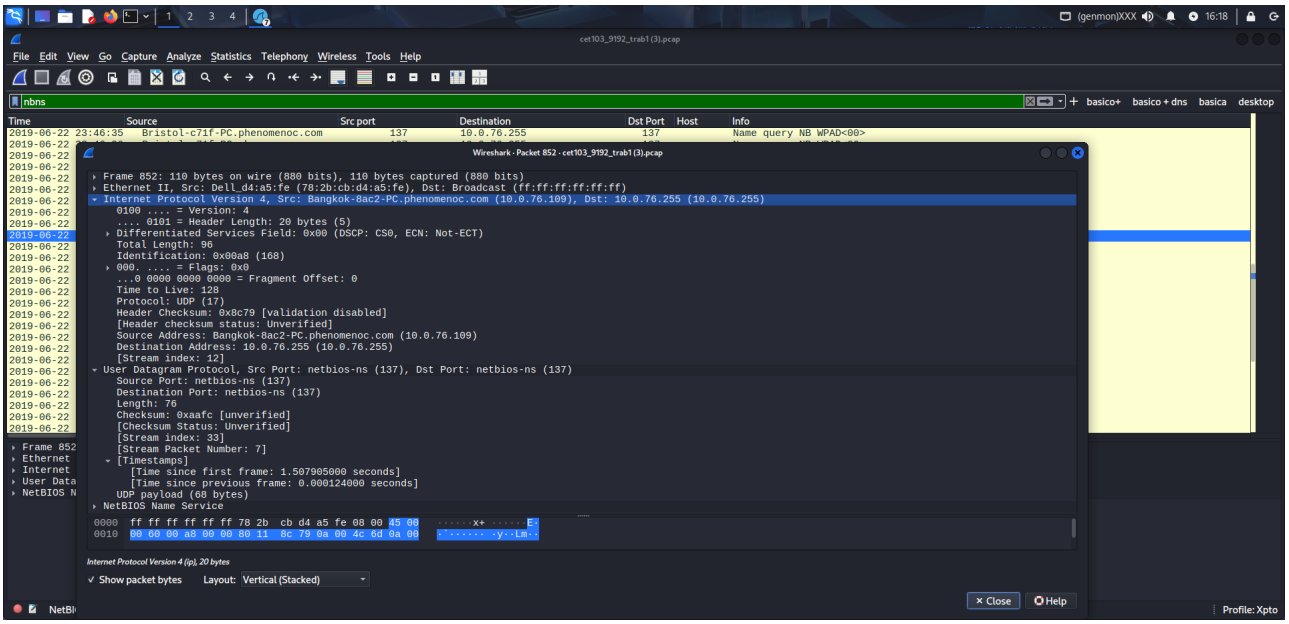


Figure 2 — Endpoint traffic and IPv4 source

6. HTTP malware delivery

HTTP requests and suspicious query strings were reviewed, including TCP stream follow-up and evidence of HTTP 200 responses. The delivery IP recorded in the exercise is 37.46.135.170.

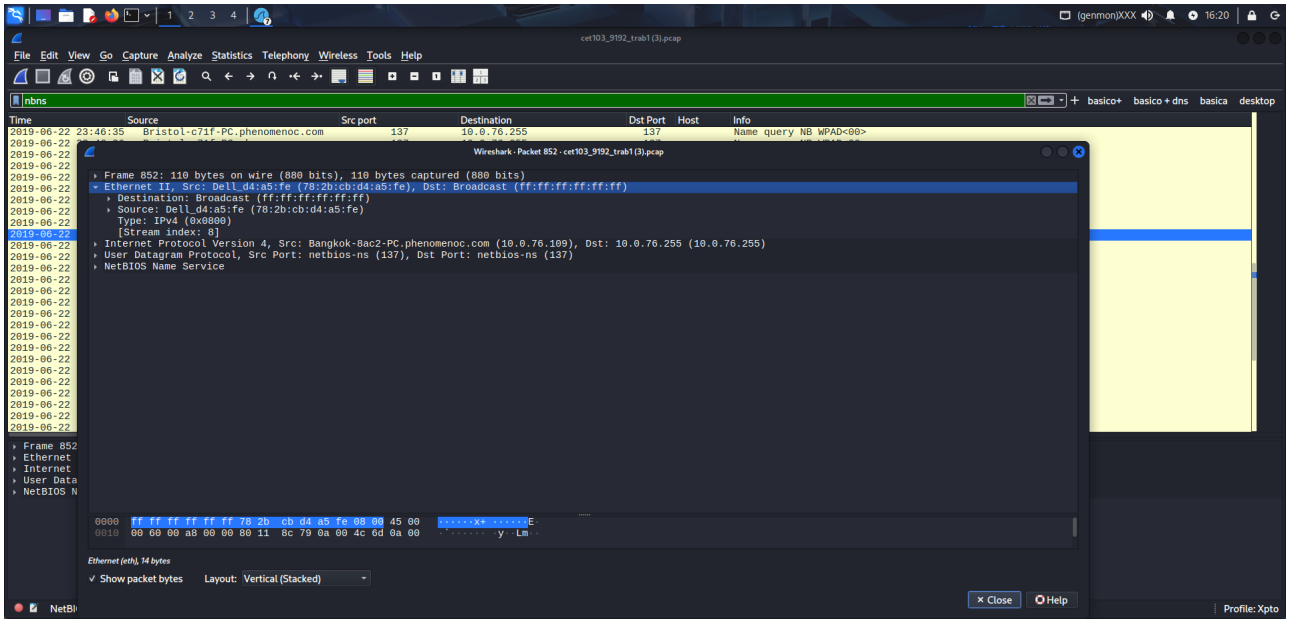


Figure 3 — Kerberos/account analysis

7. Exporting and hashing the executable

Wireshark File > Export Objects > HTTP was used to locate a candidate downloaded binary. SHA-256 was considered for sample identification and comparison with reputation information.

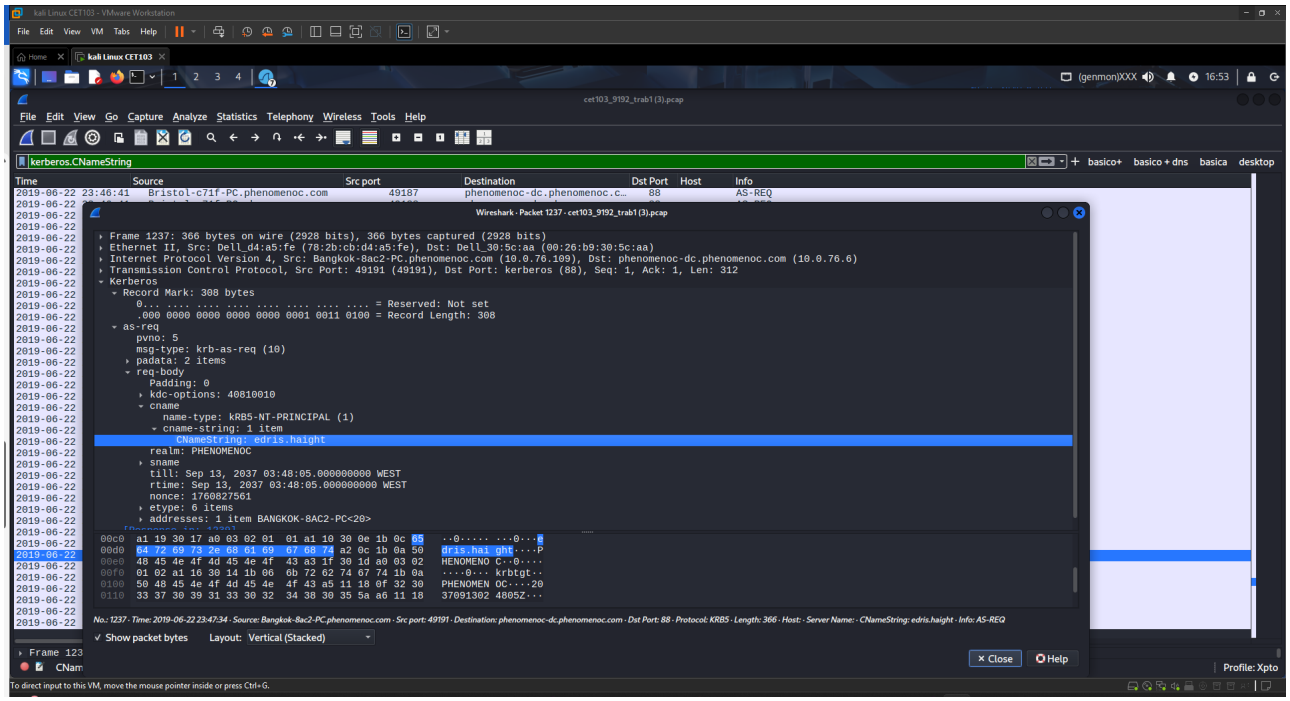


Figure 4 — HTTP delivery investigation

8. Malware classification and C2 indicators

The report associates the case with Rig Exploit Kit delivery and KPOT Stealer, and records a post-infection hostname fghjkmgru34.site and address 8.209.83.76. These are historical case indicators, not live destinations to visit.

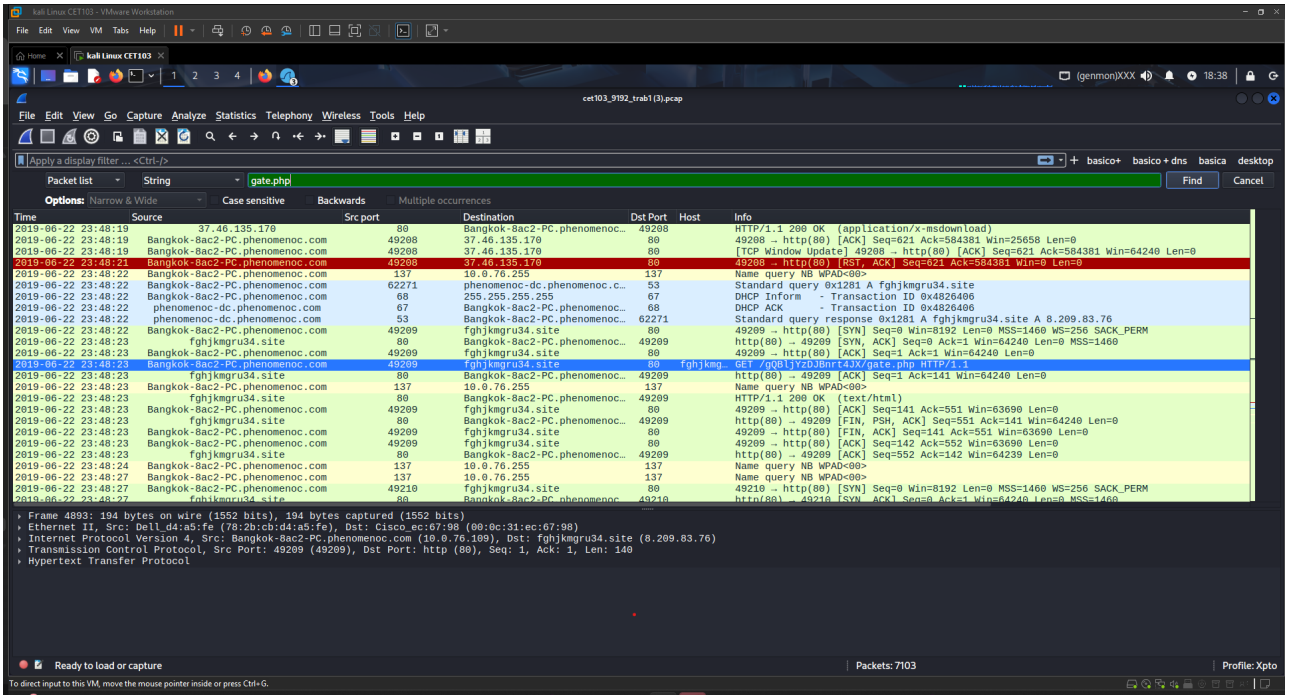


Figure 5 — HTTP stream evidence

9. Evidence limitations

The original report contains inconsistent SHA-256 values: ca5a37a5c3401ffcd1b7c98c3a22a921c013d1121fe33122e94dd81c382bf9b0 is shown in the extraction section, while 39be5610259ffade85599720ee0af31187788a00791f1e4cb0cd05ef00105eda appears in the summary. Another digest a29f4bf3c576c509dcc77245ee184cb25f38797da0bc57da96a3d78c8471ddd7 appears elsewhere. These values should not be treated as the same verified file hash without re-examining the PCAP and exported objects.

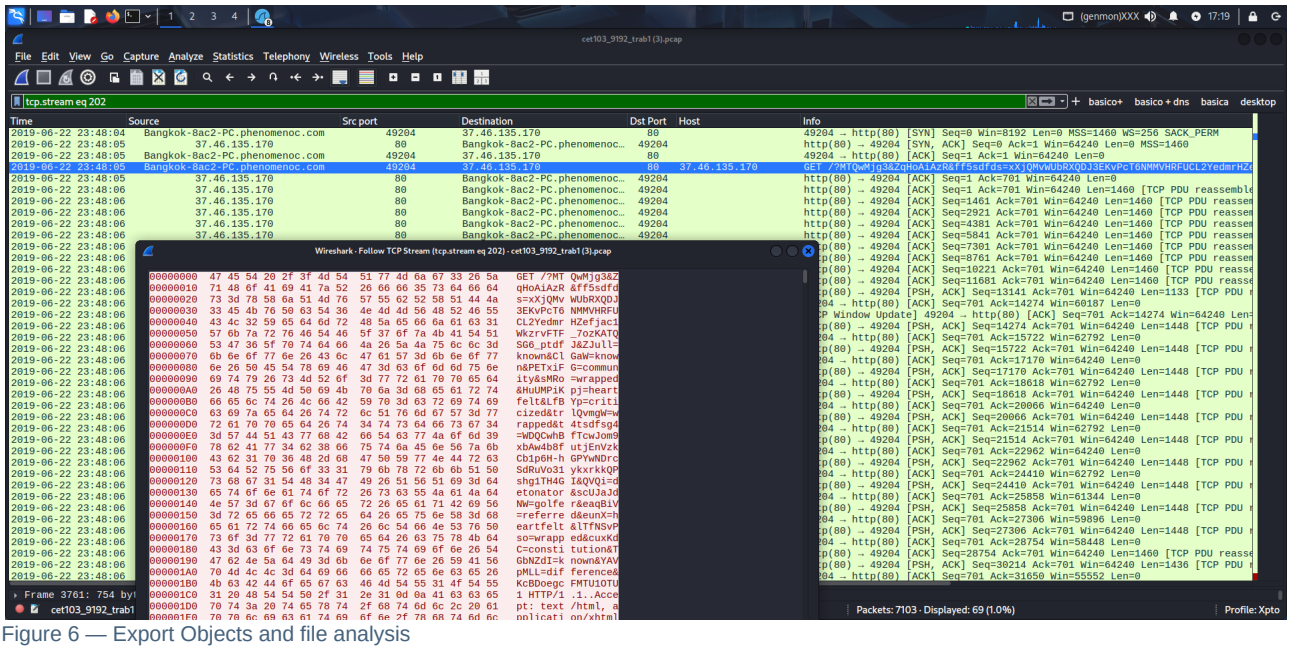


Figure 6 — Export Objects and file analysis

10. Results summary

Indicator	Value
Windows endpoint	10.0.76.109

Indicator	Value
Hostname	BANGKOK-8AC2-PC
MAC	78:2b:cb:d4:a5:fe
Windows account	edris.haight
Delivery server	37.46.135.170
Reported malware	KPOT Stealer / Rig EK context
Reported C2	fghjkmgru34.site / 8.209.83.76

11. Conclusions

The supplied evidence supports a network-forensic reconstruction of a Windows infection scenario, linking a client endpoint, suspected HTTP delivery, and subsequent suspicious communications. The case also illustrates why hash consistency, complete evidence preservation and independent validation are essential before drawing definitive conclusions.

12. References

Wireshark User Guide; NIST SP 800-61 Computer Security Incident Handling Guide (historical reference in source); ISO/IEC 27035 incident management framework.